

SHRI G.S. INSTITUTE OF TECHNOLOGY AND SCIENCE, INDORE - 3

Name : HARSH RAJ SHARMA

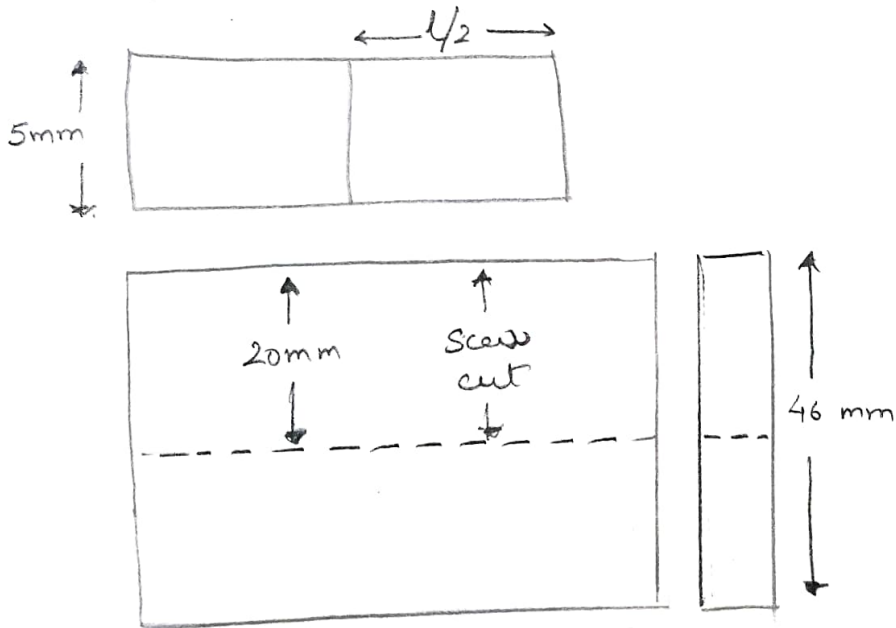
FITTING - SHOP

Roll No. : AB-16030

TITLE : To prepare job as per drawing.

OBJECTIVES : To familiarise with different marking, measuring and cutting tools.
: To learn different operation performed in the fitting shop.

DRAWING:



Raw material size :
Mild steel (flat)
60 x 40 x 6
All angles are 90°

All dia in m.m.
l → length - thread

Machine / Hand Tool	Particular	Specification
Marking Tool / Measuring Tools	Parallel jaw bench vice, hand file, Hack saw, hammer 1.) Bench vice 2.) Flat file 3.) Hand hack saw 4.) Surface plate 5.) Surface Gauge 6.) Punch 7.) Angle Plate 8.) Try Square 9.) Hammer 10.) Marking Blocks	Cast iron. Hardened high grade steel. High carbon steel. High cast iron steel. Cast iron. Forged steel.

Sequence of operation:

- 1.) Both the surface and four sides are filed at eight angles.
- 2.) Req dimensions are marked on the surface with chalks & callipers
- 3.) Marks are punched along the margin with a center punch.
- 4.) Req dimension hole is drilled out at marked center.
- 5.) Then the required cutting is done with the help of hand-saw.

Precautions:

- 1.) Job should be carefully fitted in the bench-vice, so that it can be mould.
- 2.) Avoid tapering or forming uneven surface or round surface
- 3.) Extract metal should be carefully filed, so that the punching is usable.
- 4.) Sawing should be done straight & it should not be done in punching mark.



Flat file



Hand file



Square file



Pillar file



Round file

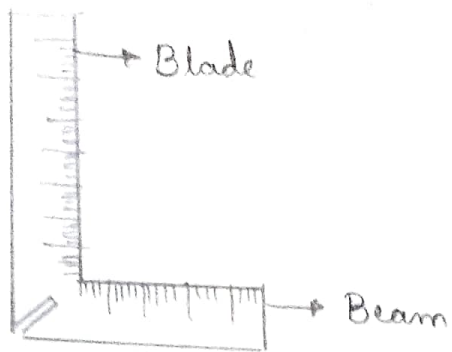


Half Round file

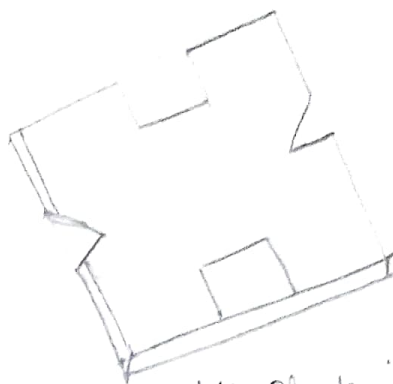


Knife edge

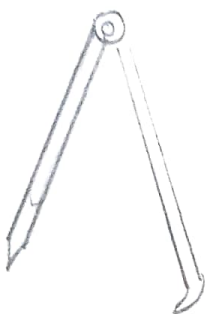




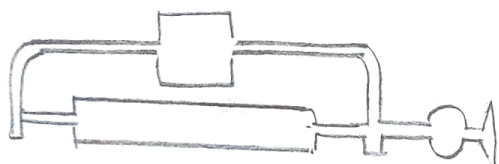
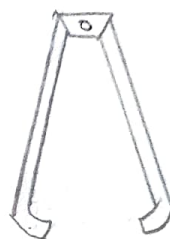
Try Square



V-Block



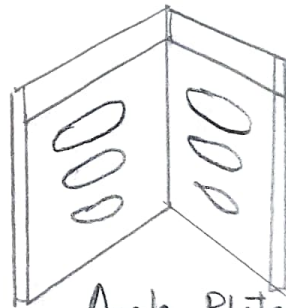
odd leg Calipers



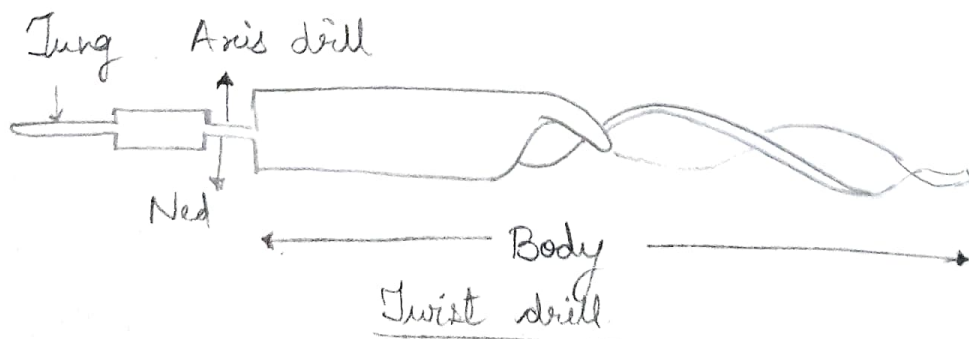
Adjustable Hack saw

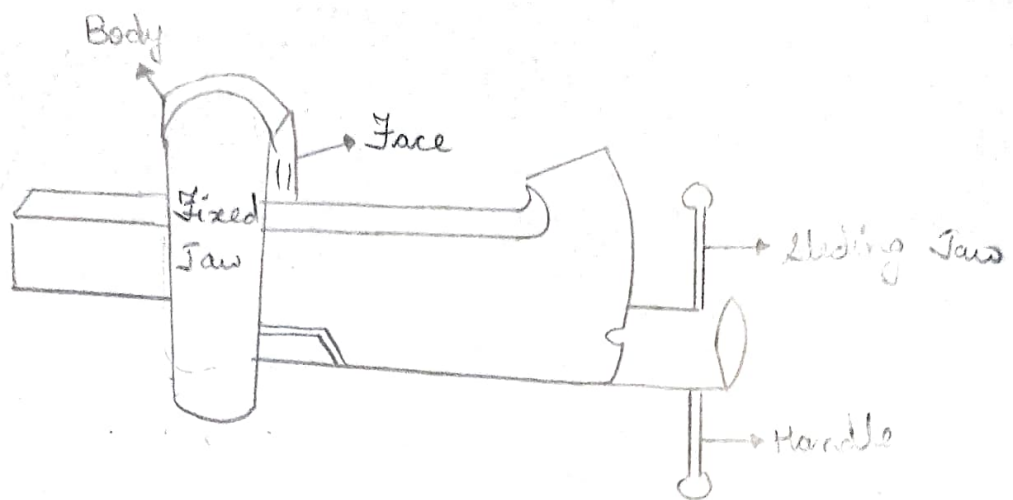


Hack saw



Angle Plate





Hole for securing to bench

Parallel Jaw Bench Vice,

QUESTIONARIES:

- 01 What are the different operation performed in the fitting shop ?
- 02 How the files are classified ?
- 03 What are the different part of the File ?
- 04 Give the specification of Hand Hacksaw blade ?
- 05 What are the different type of the vice and name the parts of bench vice specify the material also.
- 06 What are the uses of different files ?
- 07 Why should dirty and oily surface are cleaned before filing ?
- 08 What will happens if a file of smaller pitch is used than recommended.
- 09 State the reason why a hack-saw blade break ?
- 10 Write the use of following:

(i) Marking block	(ii) Angle plate
(iii) Surface plate	(iv) Centre punch and Dot punch
- 11 How tap are use for internal thread cutting ?

- 1.) (i) Clapping (ii) Filing (iii) Scraping (iv) Grinding
(v) Sawing (vi) Marking (vii) Drilling (viii) Tapering

2.) Files are classified as follows:-

- (i) Size (ii) Cut of both (a.) Single cut (b.) Double cut
(iii) Shape (a.) Flat file (b.) Hand file
(c.) Square file (d.) Pillar file (e.) Round file
(f.) Triangular file (g.) Half round file
(h.) Knife edge file

3.) Different types of file are:-

- (i) Tongs (ii) Face (iii) Point (iv) Hat
(v) Edge (vi) Teeth

4.) Specification of hand saw blade:-

Material $\Rightarrow$ High grade tool steel, hardened to HRC-61-64.

width $\Rightarrow$ 12 or 16 mm Length $\Rightarrow$ 250 mm - 300 mm

Thickness $\Rightarrow$ 0.65 mm or 0.8 mm

Holes $\Rightarrow$ Two holes of 3 mm dia at the end to accommodate prongs (Pins).

5.) Different types of vice are:-

- (i) Bench vice (ii) Leg vice (iii) Pipe vice
(iv) Hand vice (v) Pin vice (vi) Tool mark vice

6-) The use of different types of files are:-

- (i) flat file:- It is used for general work.
- (ii) Hand file:- It is used for finishing flat surface
- (iii) Square file:- It is used for filing square corner enlarging square or rectangular piece.
- (iv) filer file:- It is used for narrow work.
- (v) Round file:- It is used for filing curved surface & enlarging rounds & holes.
- (vi) Triangular file:- It is used for rectangular cuts and filling corners.
- (vii) Half round file:- It is used for round cuts & filling curved surface.
- (viii) Knife edge file:- It is used for filing narrow slot and grooves.

7-) Dirty & oily surface are cleaned & made clean before filing because dirt & oil cause the file teeth to become clogged with particles of dirt & unless the obstruction are removed, they will make deep scores across the work, also the file will become misusable.

8-) If a file of smaller pitch is used than recommended, then extra labour will be required as the work will become tougher.

It is used on a soft metal, then it will lead to scratching rather than smoothing the surface.

- 9.) (i) If workpiece is held loosely.
(ii) If there is improper tension.
(iii) If we apply improper pressure on workpiece.

10.) (i.) Marking Block:- It is used to scribe on a job in horizontal plane.

(ii.) Angle plate:- It is used in conjunction with surface plate for supporting work in perpendicular position.

(iii.) Surface plate:- It is used for testing the flatness of work itself & is also used for marking out.

(iv.) Centre punch & Dot punch:- Dot punch is used for marking small punch marks on layout lines in order to make them last longer.

11.) Taper are used for cutting internal thread in a workpiece after drilling a hole in it. These are inserted into the drilled hole one after the other and rotated with the help of tape wrench to cut the internal thread.